LAB NOTES · CAMPUS & EDGE DESIGN

Cisco Three-Tier Campus with a Layer 2 Boundary

A dual-ISP edge with IP SLA failover, route-map PAT and HSRP gateway redundancy, explained device by device and packet by packet.

TOPOLOGY	9 nodes · 2 ISPs · 6 VLANs · core, distribution and access tiers
FEATURES	IP SLA · object tracking · floating statics · route-map NAT/PAT · HSRP · Rapid-PVST · LACP · root, loop & BPDU guard · port security
PLATFORM	Cisco IOS virtual routers and switches in EVE-NG

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Appendix: device configurations A

This lab models a small enterprise campus that has outgrown a single internet circuit. Users sit in six VLANs behind an access and distribution layer, two core switches share the default gateway, and one edge router connects to two service providers. If the primary provider stops delivering traffic, the edge moves everything to the backup within seconds and moves it back once the primary is healthy again.

Each section explains one mechanism: what the configuration says, why each line is there, and what you should see when you verify it. Command output is described, not pasted, so run the commands in section 11 on your own build and compare.

The design at a glance

Three switching tiers carry six VLANs from the access port to the core. The core is where Layer 2 ends: every VLAN has its gateway on a pair of core SVIs, and everything north of the core is routed.

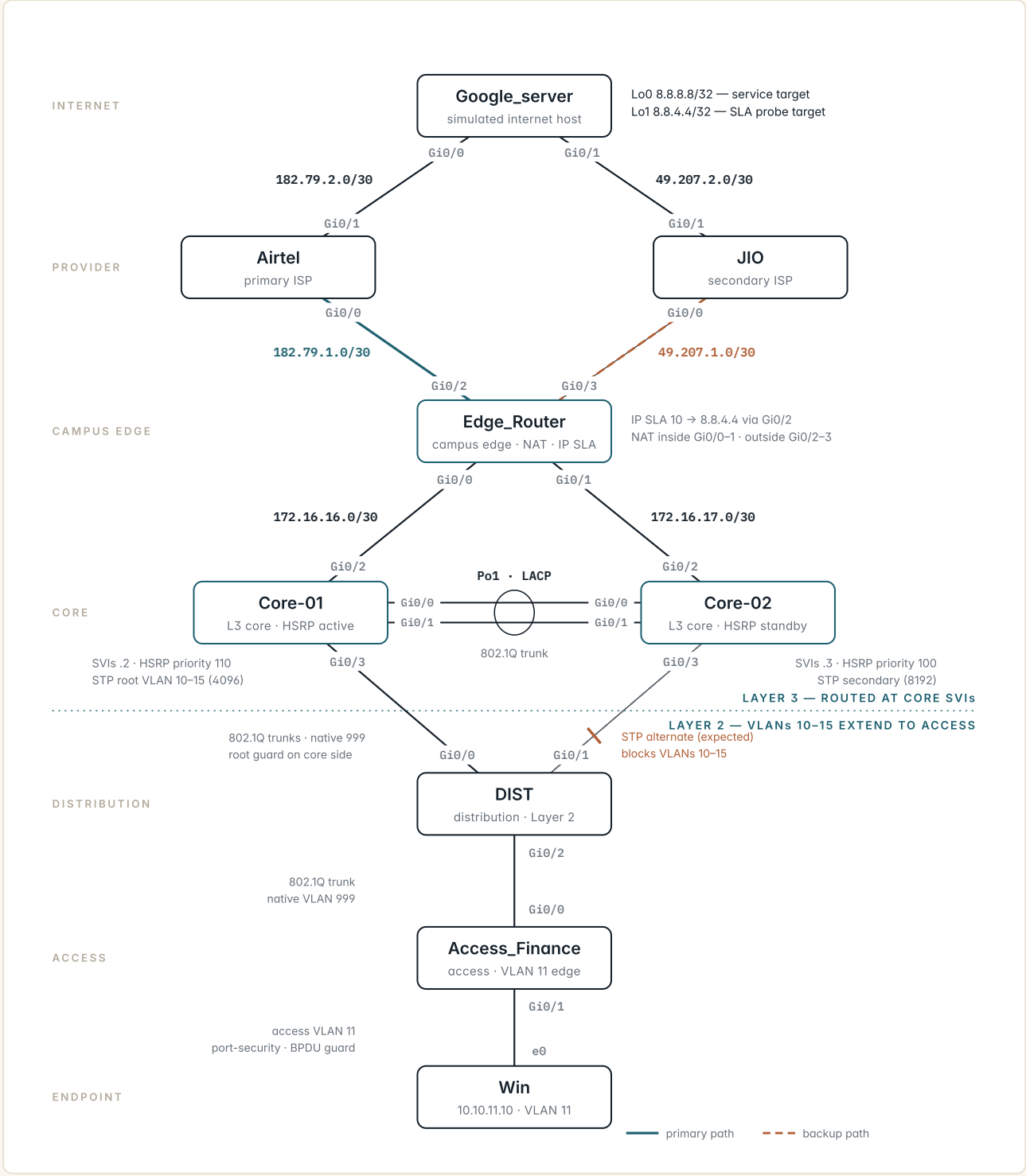


Figure 1. Lab topology. Port labels sit at each end of a link; /30 subnets are shown on routed links. The STP alternate port on DIST Gi0/1 is the expected result of root placement (section 08), not a configured block.

How the pieces fit

EDGE AND INTERNET

Edge_Router has one /30 to each core and one /30 to each provider. A single IP SLA probe tests the Airtel path end to end. While it succeeds, the default route points at Airtel; when it fails, a floating default via JIO takes over. Two route-maps make sure the PAT address always belongs to whichever ISP the packet actually leaves through.

CORE

Core-01 and Core-02 run HSRP for VLANs 10 to 15. Core-01 is active for every group and is also the spanning-tree root for the same VLANs, so the Layer 2 and Layer 3 forwarding paths line up. Both cores track their own uplink to the edge and give up the active role if it fails.

DISTRIBUTION AND ACCESS

DIST is a pure Layer 2 switch with a management address in VLAN 10. It is dual-homed to both cores, which creates a loop that Rapid-PVST breaks by blocking the uplink toward Core-02. Access_Finance has one uplink and one user port in VLAN 11, protected by port security and BPDU guard.

WHY A LAYER 2 BOUNDARY AT THE CORE

Keeping distribution at Layer 2 lets any VLAN appear on any access switch without re-addressing, at the cost of a larger spanning-tree domain. The alternative, routed links from distribution upward, removes STP from the uplinks but pins each subnet to one distribution block. This lab deliberately shows the first model and its guard rails.

VLAN	NAME	SUBNET	GATEWAY (VIP)	CORE-01	CORE-02
`10`	Device management	`10.10.10.0/24`	`10.10.10.1`	`.2`	`.3`
`11`	Data	`10.10.11.0/24`	`10.10.11.1`	`.2`	`.3`
`12`	Voice	`10.10.12.0/24`	`10.10.12.1`	`.2`	`.3`
`13`	Guest	`10.10.13.0/24`	`10.10.13.1`	`.2`	`.3`
`14`	BYOD	`10.10.14.0/24`	`10.10.14.1`	`.2`	`.3`
`15`	Quarantine	`10.10.15.0/24`	`10.10.15.1`	`.2`	`.3`
`999`	Native VLAN on access-side trunks (no SVI)	—	—	—	—

Table 1. VLAN plan. VLAN 15 is described as "Corentine Vlan" in the configs; it is used here as a quarantine VLAN.

Port matrix and addressing

Every link in the lab, read from interface configurations and the topology. Each routed link is a /30 with one usable address per side.

Routed point-to-point links

#	DEVICE A	PORT	IP ADDRESS	DEVICE B	PORT	IP ADDRESS	SUBNET
1	Edge_Router	`Gi0/2`	`182.79.1.2`	Airtel	`Gi0/0`	`182.79.1.1`	`182.79.1.0/30`
2	Edge_Router	`Gi0/3`	`49.207.1.2`	JIO	`Gi0/0`	`49.207.1.1`	`49.207.1.0/30`
3	Airtel	`Gi0/1`	`182.79.2.1`	Google_server	`Gi0/0`	`182.79.2.2`	`182.79.2.0/30`
4	JIO	`Gi0/1`	`49.207.2.1`	Google_server	`Gi0/1`	`49.207.2.2`	`49.207.2.0/30`
5	Edge_Router	`Gi0/0`	`172.16.16.1`	Core-01	`Gi0/2`	`172.16.16.2`	`172.16.16.0/30`
6	Edge_Router	`Gi0/1`	`172.16.17.1`	Core-02	`Gi0/2`	`172.16.17.2`	`172.16.17.0/30`

Switched links

#	DEVICE A	PORT	DEVICE B	PORT	MODE	NOTES
7	Core-01	`Gi0/0`	Core-02	`Gi0/0`	Trunk, Po1	LACP active both sides
8	Core-01	`Gi0/1`	Core-02	`Gi0/1`	Trunk, Po1	LACP active both sides
9	Core-01	`Gi0/3`	DIST	`Gi0/0`	Trunk	Native 999 · root guard on core · DIST root port
10	Core-02	`Gi0/3`	DIST	`Gi0/1`	Trunk	Native 999 · root guard on core · DIST alternate
11	DIST	`Gi0/2`	Access_Finance	`Gi0/0`	Trunk	Native 999
12	Access_Finance	`Gi0/1`	Win	`e0`	Access	VLAN 11 · port security · PortFast · BPDU guard

Host and loopback addresses

DEVICE	INTERFACE	ADDRESS	ROLE
Google_server	`Lo0`	`8.8.8.8/32`	Service target
Google_server	`Lo1`	`8.8.4.4/32`	IP SLA probe target
DIST	`Vlan10`	`10.10.10.10/24`	Management
Access_Financ	`Vlan10`	`10.10.10.15/24`	Management
Win	`e0`	`10.10.11.10/24`	User endpoint

Spare ports (configured, not cabled)

DEVICE	PORT	STATE IN CONFIG
Core-01	`Gi1/0`	Trunk, native 999, root guard
Core-02	`Gi1/0`	Trunk, native 999, root guard
DIST	`Gi0/3`	Trunk, native 999
Access_Finance	`Gi0/2`	Trunk, native 999

These ports are configured as trunks but nothing is connected to them. Shutting them down is covered in the review notes.

Edge routing: a tracked primary and a floating backup

The edge router has no routing protocol. Two static default routes decide where internet traffic goes, and an object track decides which of them is allowed into the routing table.

```
ip route 0.0.0.0 0.0.0.0 182.79.1.1 track 1
ip route 0.0.0.0 0.0.0.0 49.207.1.1 10
```

`0.0.0.0 0.0.0.0`	The default route: matches any destination that has no more specific entry.
`182.79.1.1 track 1`	Next hop is Airtel. The route is installed only while track object 1 is *Up*. When the track goes Down, IOS withdraws the route even though the interface is still up.
`49.207.1.1 10`	Next hop is JIO with administrative distance 10. A static route defaults to AD 1, so this one loses to the Airtel route whenever both are valid. It sits in reserve and installs only when the Airtel route disappears. That is a *floating static*.

Why distance 10 and not 2

Any value above 1 makes the JIO route float. Leaving a gap means a third path, or a dynamic protocol with a low default distance, can be slotted in later without renumbering everything. Just keep the backup's distance below the protocol you never want it to beat.

Why interface tracking would not be enough

A static route already disappears if its outgoing interface goes down. The weak spot is a failure further away: the Airtel handoff stays up, but Airtel's link to the internet is broken. The edge would keep sending traffic into a dead path. Tracking the result of an end-to-end probe, rather than the interface, closes that gap. Section 04 builds the probe.

UNDER REVIEW

The exported Edge configuration contains three more static lines with next hops 8.8.8.8 and 4.4.4.4, plus a host route to 1.1.1.1. Neither 4.4.4.4 nor 1.1.1.1 exists in this topology, and 8.8.8.8 is a far-end host rather than a next hop. They do not change the behaviour described here and are left out of the walkthrough until confirmed. See Review notes, item 1.

04 FAILURE DETECTION

IP SLA and object tracking: the failover trigger

IP SLA sends a ping every five seconds to a dedicated address beyond Airtel. A track object turns the stream of results into a single Up or Down state, with dampening so a single lost packet does not trigger a failover.

```
ip sla 10
 icmp-echo 8.8.4.4 source-interface GigabitEthernet0/2
 frequency 5
ip sla schedule 10 life forever start-time now
!
track 1 ip sla 10 reachability
 delay down 10 up 15
!
ip route 8.8.4.4 255.255.255.255 182.79.1.1
```

icmp-echo 8.8.4.4	The probe target is Loopback1 on Google_server, two hops beyond the edge. A reply proves Airtel's access link, its routing and its upstream link all work. Pinging 182.79.1.1 would only prove the first hop.
source-interface Gi0/2	Probes leave with source 182.79.1.2, an Airtel address. Google_server returns traffic for 182.79.1.0/30 via Airtel, so the reply comes back the same way. Router-originated traffic is not translated by the inside-source NAT rules.
frequency 5	One probe every 5 seconds. The timeout is not set, so the IOS default of 5000 ms applies. Confirm it with `show ip sla configuration 10`.
schedule … life forever start-time now	An IP SLA operation does nothing until it is scheduled. This starts it immediately and runs it permanently.
track 1 ip sla 10 reachability	The track is Up while the operation's return code is OK and Down when probes fail or time out.
delay down 10 up 15	The state must persist before the track changes: 10 seconds of failures before going Down, 15 seconds of success before coming back Up. Recovery is deliberately slower so that a flapping primary does not drag traffic back and forth.
ip route 8.8.4.4/32 via 182.79.1.1	Pins the probe to Airtel. Without it, once the default route moves to JIO the probe would follow it, and the track would start reporting JIO's health as if it were Airtel's. This host route is intentionally not tracked.

FAILOVER TIMING WITH THESE VALUES

Detect Up to 5 s until the next probe, up to 5 s for it to time out, then 10 s of delay. Roughly **15 to 20 s** from failure to the backup route being installed.

Recover Up to 5 s for the first good probe, then 15 s of delay. Roughly **15 to 20 s** after the primary path is healthy again.

Calculated from the configured values, not measured. Time a real failover in the lab with a continuous ping from Win.

PRODUCTION NOTE

Here JIO has no route to 8.8.4.4, so the probe cannot succeed through the backup even if the pinned route disappears. A real backup ISP would route it; production designs add `ip route 8.8.4.4 255.255.255.255 Null0 250` so the probe is dropped, not rerouted.

Dual-ISP PAT with route-maps

Each provider only routes return traffic to its own addresses. A packet that leaves through JIO carrying Airtel's address will never get a reply. Route-maps let one router run two PAT rules and pick the right one based on where the packet is actually going.

```
interface GigabitEthernet0/0
 ip nat inside
interface GigabitEthernet0/1
 ip nat inside
interface GigabitEthernet0/2
 ip nat outside
interface GigabitEthernet0/3
 ip nat outside
!
ip access-list standard LAN_NETWORKS
 permit 10.10.0.0 0.0.255.255
```

```
route-map RM_NAT_AIRTEL permit 10
 match ip address LAN_NETWORKS
 match interface GigabitEthernet0/2
!
route-map RM_NAT_JIO permit 10
 match ip address LAN_NETWORKS
 match interface GigabitEthernet0/3
```

```
ip nat inside source route-map RM_NAT_AIRTEL interface GigabitEthernet0/2 overload
ip nat inside source route-map RM_NAT_JIO interface GigabitEthernet0/3 overload
```

ip nat inside / outside	Marks the direction. Both core-facing links are inside because either core may be forwarding user traffic; both ISP links are outside.
LAN_NETWORKS	Matches every 10.10.x.x source. All six VLANs, including management, are translated.
match ip address + match interface	Two match lines in one route-map sequence are combined with AND. The source must be internal *and* the packet must be leaving through that ISP's interface.
match interface Gi0/2	In a NAT route-map, this matches the egress interface the routing table chose. That is what ties the translation to the path.
interface Gi0/2 overload	Translate to whatever address Gi0/2 currently holds and multiplex all inside hosts onto it using source ports. That is PAT.

Order of operations

For traffic going from inside to outside, IOS makes the routing decision before it translates. By the time NAT evaluates the route-maps it already knows the egress interface, so `match interface` has something real to match. Change the default route and the next new flow automatically picks up the other ISP's address. No NAT change is needed during failover.

What happens to existing sessions

A translation is created when a flow starts and stays in the table until it ages out. After a failover, entries built on Airtel's address do not move to JIO. Those sessions stall and the applications reconnect, and new flows are translated to 49.207.1.2. `clear ip nat translation *` removes stale entries at once, but it also drops every active translation, so treat it as a disruptive command.

CHECK IN THE LAB

The edge also sets `ip nat translation timeout 30`. Under PAT, per-protocol timers (TCP, UDP, ICMP) also govern how long individual entries live, so confirm the effective aging with `show ip nat translations verbose` before relying on the 30-second value in an explanation.

Return-path routing to the campus

The cores know how to reach the edge through their own default routes. The edge also needs to know how to reach the campus subnets, so it holds two static routes for each VLAN: a primary via Core-01 and a floating backup via Core-02.

```
! Edge_Router (pattern repeats for 10.10.10.0 through 10.10.15.0)
ip route 10.10.11.0 255.255.255.0 172.16.16.2
ip route 10.10.11.0 255.255.255.0 172.16.17.2 250
!
! Core-01                                  ! Core-02
ip route 0.0.0.0 0.0.0.0 172.16.16.1       ip route 0.0.0.0 0.0.0.0 172.16.17.1
```

Why the primary points at Core-01

Core-01 is HSRP active and STP root, so user traffic leaves the campus through Core-01. Sending return traffic back through the same switch keeps the path symmetric. If it came back through Core-02 instead, every reply would cross Po1 to reach the active forwarding path, which works but wastes the inter-core link and complicates troubleshooting.

Two mechanisms, one failure

Watch what happens when the Edge–Core-01 link fails. Both sides react on their own, and they end up in agreement:

DEVICE	DETECTS	ACTION
Edge_Router	Gi0/0 down, so 172.16.16.0/30 leaves the table	Statics via 172.16.16.2 can no longer resolve; the AD 250 routes via 172.16.17.2 install
Core-01	Track 1 (Gi0/2 line protocol) goes Down	HSRP priority drops from 110 to 90 on every group
Core-02	Receives hellos with priority 90	Preempts with priority 100 and becomes active for VLANs 10–15

Outbound traffic now leaves through Core-02 and return traffic arrives at Core-02, so the path stays symmetric with no routing protocol involved.

LIMITATION

Both mechanisms rely on a link going down. If Core-01 stopped forwarding while its uplink stayed up, the edge would keep using the stale primary route. That is rare with direct cabling, but common when a carrier device or a transceiver sits in the middle. The usual answers are an IP SLA track toward each core's /30 address, or running OSPF between the edge and the cores.

Gateway redundancy: HSRP with tracking

Hosts point at a single gateway address that neither core owns. HSRP decides which core answers for it, and object tracking makes that decision aware of the uplink.

CORE-01

```
track 1 interface GigabitEthernet0/2 line-
protocol
!
interface Vlan11
 ip address 10.10.11.2 255.255.255.0
 standby 11 ip 10.10.11.1
 standby 11 priority 110
 standby 11 preempt
 standby 11 authentication md5 key-string
SecGroup11
 standby 11 track 1 decrement 20
```

CORE-02

```
track 1 interface GigabitEthernet0/2 line-
protocol
!
interface Vlan11
 ip address 10.10.11.3 255.255.255.0
 standby 11 ip 10.10.11.1
 !  priority not set: default 100
 standby 11 preempt
 standby 11 authentication md5 key-string
SecGroup11
 standby 11 track 1 decrement 20
```

standby 11 ip 10.10.11.1	The virtual gateway address. Group number 11 matches the VLAN for readability. HSRP version 1 is in use (no `standby version 2`), so the virtual MAC is 0000.0c07.ac0b, where 0b is group 11 in hex.
priority 110 / default 100	The higher priority becomes active. Core-01 wins every group in normal operation.
preempt	Lets a router with a higher priority take over from the current active router. It is needed on *both* cores (see below).
authentication md5	Both peers must share the key per group; hellos with a wrong key are ignored. This stops a rogue device from claiming the gateway.
track 1 decrement 20	If Gi0/2 goes down, subtract 20 from this router's priority. Track 1 on the cores is a different object from track 1 on the edge; track numbers are local to each device.

Why both cores need preempt

When Core-01 loses its uplink it drops to 90, but a lower priority alone does not hand over the active role. Core-02 takes over only because it is allowed to preempt. Without preempt on Core-02, Core-01 would stay active and send traffic across Po1 toward a core that cannot reach the edge. The preempt on Core-01 does the opposite job: it lets Core-01 reclaim the role once its uplink returns and its priority goes back to 110.

CONDITION	CORE-01	CORE-02	ACTIVE GATEWAY
Normal	110	100	Core-01
Core-01 uplink down	90	100	Core-02 (preempts)
Core-02 uplink down	110	80	Core-01 (no change)
Core-01 uplink restored	110	100	Core-01 (preempts back)

Timers are at the defaults (hello 3 s, hold 10 s). A tracked failover takes effect at the next hello carrying the new priority. A complete Core-01 failure is noticed only when the hold timer expires.

The Layer 2 domain: STP, EtherChannel and edge protection

Because VLANs stretch from the access switch to both cores, the design has a physical loop: DIST, Core-01 and Core-02 form a triangle. Spanning tree breaks it, and a set of guard features keeps the result predictable.

Root placement, aligned with HSRP

```
! Core-01                                  ! Core-02
spanning-tree mode rapid-pvst              spanning-tree mode rapid-pvst
spanning-tree vlan 10-15 priority 4096     spanning-tree vlan 10-15 priority 8192
```

Core-01 is root for VLANs 10–15 and Core-02 is the backup root. DIST hears a BPDU from each core and picks the cheaper path to the root. Directly through Gi0/0 the cost is 4, since a 1 Gb/s link costs 4 under the short path-cost method. Through Gi0/1 it costs 4 plus Po1's cost of 3, so 7. DIST Gi0/0 becomes the root port. On the DIST–Core-02 segment, Core-02 advertises the better cost, so Core-02 Gi0/3 is designated and **DIST Gi0/1 becomes the alternate (blocking) port**. Frames from Access_Finance therefore go straight to Core-01, which is also the active gateway. No frame needs to cross Po1 in normal operation. These costs assume 1 Gb/s interfaces and the short method; confirm the roles with `show spanning-tree vlan 11` on DIST.

EtherChannel between the cores

Gi0/0 and Gi0/1 on each core are bundled into Port-channel1 with `channel-group 1 mode active`, which is LACP. Spanning tree sees one logical link, so neither member is blocked, and LACP negotiation keeps a miscabled or one-way member out of the bundle. Static `mode on` does not offer that protection.

Guard rails

FEATURE	WHERE	WHAT IT PREVENTS
`spanning-tree guard root`	Core-01 and Core-02 Gi0/3	A switch below the core announcing a better bridge ID and taking the root role. The port goes root-inconsistent instead of accepting it.
`spanning-tree loopguard default`	Both cores, global	A root or alternate port moving to forwarding because BPDUs stopped arriving, for example on a one-way link. The port goes loop-inconsistent instead.
`spanning-tree bpduguard enable`	Access_Finance Gi0/1	A switch plugged into a user port. Any BPDU err-disables the port.
`spanning-tree portfast edge`	Access_Finance Gi0/1	Waiting through STP states on a host port. The port forwards immediately.
`switchport trunk native vlan 999`	All access-side trunks	Untagged frames landing in a VLAN with hosts or an SVI, and double-tagging VLAN hopping. Must match on both ends of each trunk.
`switchport port-security … sticky`	Access_Finance Gi0/1	Any device other than the learned MAC 0050.7966.6808 on that port. Defaults apply: one MAC, and a violation shuts the port down.

Root guard and loop guard interact on the same port; confirm which is active on the core downlinks with `show spanning-tree interface Gi0/3 detail`. DIST's `spanning-tree portfast edge bpduguard default` only affects PortFast edge ports, and DIST has none yet.

09 END TO END

Packet walk: normal path and failover

One ICMP echo from Win to 8.8.8.8, hop by hop, first with both providers healthy and then after the Airtel path has failed.

Normal operation

	WHERE	WHAT HAPPENS
1	**Win**	Destination 8.8.8.8 is off-subnet, so Win ARPs for its gateway `10.10.11.1`. Core-01, the HSRP active router, replies with the virtual MAC `0000.0c07.ac0b`.
2	**Access_Finance**	The frame arrives on `Gi0/1` (access, VLAN 11) and leaves the `Gi0/0` trunk tagged VLAN 11.
3	**DIST**	In on `Gi0/2`, out on root port `Gi0/0` toward Core-01. `Gi0/1` is alternate and forwards nothing.
4	**Core-01**	`Vlan11` receives the frame and routes the packet. The default route sends it to `172.16.16.1` out `Gi0/2`.
5	**Edge_Router**	In on `Gi0/0` (NAT inside). Track 1 is Up, so the default points to `182.79.1.1` and the egress is `Gi0/2`.
6	**Edge_Router**	`RM_NAT_AIRTEL` matches (source in LAN_NETWORKS, egress Gi0/2). Source `10.10.11.10` is translated to `182.79.1.2` with a new source port or ICMP identifier.
7	**Airtel**	Host route `8.8.8.8/32` via `182.79.2.2` delivers the packet to Google_server Lo0.
8	**Return**	Google_server sends the reply to `182.79.1.2` via `182.79.2.1`. Edge un-translates it to `10.10.11.10` and routes `10.10.11.0/24` via `172.16.16.2` back to Core-01, then DIST and Access to Win.

After the Airtel path fails

Assume Airtel's link to Google_server breaks. The edge's Gi0/2 stays up, so only the probe can notice.

	WHERE	WHAT CHANGES
1	**Edge_Router**	Probes to `8.8.4.4` time out. After 10 s of failure, track 1 goes Down.
2	**Edge_Router**	The tracked default via `182.79.1.1` is withdrawn. The floating default via `49.207.1.1` (AD 10) installs.
3	**Edge_Router**	New flows leave via `Gi0/3`. `RM_NAT_JIO` matches and the source becomes `49.207.1.2`.
4	**JIO**	Host route `8.8.8.8/32` via `49.207.2.2` delivers the packet.
5	**Return**	Google_server routes `49.207.1.0/30` via `49.207.2.1`, so the reply returns through JIO and is un-translated on the edge as before.
6	**Campus**	Nothing changes inside the campus: same gateway, same core, same spanning-tree path.

Note what JIO does *not* carry: it has no route to 8.8.4.4. The service target stays reachable over the backup, but the probe target does not. The track keeps reporting on Airtel alone and cannot be fooled by the backup path.

Failure scenarios

Expected behaviour for each failure, derived from the configuration. Each row doubles as a lab exercise: break the component and check that the result matches.

FAILURE	WHAT DETECTS IT	WHAT CHANGES	EXPECTED IMPACT
Airtel upstream fails (Airtel–Google link)	IP SLA 10 times out; track 1 Down after 10 s	Default moves to JIO; NAT uses RM_NAT_JIO	About 15–20 s for new flows; existing sessions reconnect
Edge Gi0/2 or Airtel handoff down	Connected route lost; pinned probe route and tracked default fall out	Same as above	Similar; the backup route installs once the primary is gone
Airtel path restored	Probes succeed; track Up after 15 s	Primary default returns; JIO route floats again	New flows move back to Airtel; JIO translations age out
Edge–Core-01 link down	Edge loses 172.16.16.0/30; Core-01 track 1 Down	Edge return routes float to Core-02; Core-02 becomes HSRP active	Seconds, within the HSRP hello interval
Core-01 fails completely	HSRP hold timer; RSTP on DIST; Edge Gi0/0 down	Core-02 active for all groups and root for VLANs 10–15	Up to about 10 s (HSRP hold time)
DIST Gi0/0 fails	RSTP on DIST	Gi0/1 alternate becomes root port	Brief; traffic now runs DIST → Core-02 → Po1 → Core-01
Switch plugged into Win's port	BPDU guard	Access_Finance Gi0/1 err-disabled	Port down until recovered
Different device on Win's port	Port security (1 sticky MAC)	Violation shutdown, err-disabled	Port down until recovered
Rogue switch claims STP root below the core	Root guard on core Gi0/3	Port goes root-inconsistent	Root stays at Core-01

WORTH NOTICING

When DIST loses its uplink to Core-01 the gateway does not move, because Core-01 is still healthy and still HSRP active. Traffic simply takes the longer Layer 2 path through Core-02 and across Po1. That is exactly why the inter-core link is a two-member EtherChannel rather than a single cable.

Verification checklist

Run these after building the lab and again during each failure test. Output formats differ between IOS images, so this list says what to look for rather than reproducing output.

DEVICE	COMMAND	LOOK FOR
Edge	`show ip sla configuration 10`	Target 8.8.4.4, source Gi0/2, frequency 5 s, timeout value
Edge	`show ip sla statistics 10`	Latest return code OK; success counter increasing
Edge	`show track 1`	State Up; delay down 10, up 15; number of state changes
Edge	`show ip route 0.0.0.0`	Next hop 182.79.1.1 while healthy, 49.207.1.1 after failover
Edge	`show ip route static`	One default installed at a time; 10.10.x.0/24 via 172.16.16.2
Edge	`show ip nat translations`	Inside local 10.10.11.10 mapped to 182.79.1.2 (or 49.207.1.2 after failover)
Edge	`show ip nat statistics`	Inside and outside interfaces listed; hits increasing; both route-maps referenced
Edge	`show route-map`	Both maps present with policy match counters
Cores	`show standby brief`	Core-01 Active, priority 110; Core-02 Standby, priority 100, for groups 10–15
Cores	`show track 1`	Gi0/2 line protocol Up
Cores	`show etherchannel summary`	Po1 in use, both members bundled
Cores	`show lacp neighbor`	Partner seen on both members
Cores	`show spanning-tree root`	Core-01 is root for VLANs 10–15
Cores	`show spanning-tree inconsistentports`	Empty in normal operation
DIST	`show spanning-tree vlan 11`	Gi0/0 Root FWD, Gi0/1 Altn BLK
All switches	`show interfaces trunk`	Native VLAN 999 on access-side trunks; VLANs 10–15 allowed and forwarding
Access	`show port-security interface Gi0/1`	Secure-up, maximum 1, sticky MAC 0050.7966.6808
Win	`ping 8.8.8.8 -t` / `tracert 8.8.8.8`	Replies; path via 10.10.11.1 and the edge; time the gap during failover

LAB ONLY

`debug ip nat` and `debug ip sla trace` are useful in EVE-NG but can load the CPU on a busy production router. Scope them with an ACL or use them only in a maintenance window. To test failover, shut Airtel's Gi0/1 on the Airtel router rather than the edge interface; that reproduces the far-side failure that only IP SLA can see.

Review notes

Pending confirmation

1. **Extra static routes on Edge_Router.** `ip route 0.0.0.0 0.0.0.0 8.8.8.8 track 1`, `ip route 0.0.0.0 0.0.0.0 4.4.4.4 10` and `ip route 1.1.1.1 255.255.255.255 8.8.8.8`. 4.4.4.4 and 1.1.1.1 do not exist in the topology, and 8.8.8.8 can only be reached through a default route, so these next hops add nothing to the design. If they are leftovers from testing, remove them. If they are intentional, capture `show ip route 0.0.0.0` with them present before documenting their effect.
2. **NAT translation timeout.** Confirm how `ip nat translation timeout 30` interacts with the per-protocol PAT timers on this image (section 05).

Production hardening

The lab works as built. These are the changes a production deployment of the same design would normally add.

3. **Protect the probe route.** Add a Null0 backup for 8.8.4.4 at a high distance (section 04).
4. **Detect more than link loss toward the cores.** Track each core's /30 address with IP SLA, or run OSPF between the edge and the cores (section 06).
5. **Prune trunks.** Allow only the VLANs in use, for example `switchport trunk allowed vlan 10-15,999`, rather than all VLANs.
6. **Pin the root for every VLAN.** Priorities are set only for VLANs 10–15. VLAN 1 and 999 elect a root by lowest MAC, which could be any switch.
7. **Shut spare ports.** Core-01 Gi1/0, Core-02 Gi1/0, DIST Gi0/3 and Access_Finance Gi0/2 are live trunks with nothing attached.
8. **Management plane.** Disable `ip http server` on the switches; replace type 5 (MD5-based) secrets with type 9 (`algorithm-type scrypt`); restrict VTY access to the management VLAN with an access-class.
9. **Domain name.** `ip domain-name Hydent@1234` is a valid string to IOS but not a valid DNS name. Use something like `hydent.lab`. It is also part of the RSA key name SSH relies on.
10. **VLAN definitions.** VLANs 10–15 and 999 are not in the running configurations shown. In VTP server or client mode they live in `vlan.dat`. Make sure they exist on every switch, or the trunks will not carry them.
11. **NAT scope.** LAN_NETWORKS translates all of 10.10.0.0/16, including device management in VLAN 10. Narrow it if management addresses should not reach the internet.

A APPENDIX

Device configurations

From the lab export. Platform defaults (duplex, speed, media-type, negotiation, boot markers and empty `!` lines) are omitted and static routes are grouped for readability. The unapplied `MGMT_ONLY_ACL` is removed, and uncabled ports are described as spare.

Edge_Router

Campus edge router

```
hostname Edge_Router
!
track 1 ip sla 10 reachability
 delay down 10 up 15
!
interface GigabitEthernet0/0
 description Downlink to Switch1 Gi0/2
 ip address 172.16.16.1 255.255.255.252
 ip nat inside
 ip virtual-reassembly in
!
interface GigabitEthernet0/1
 description Downlink to Switch2 Gi0/2
 ip address 172.16.17.1 255.255.255.252
 ip nat inside
 ip virtual-reassembly in
!
interface GigabitEthernet0/2
 description UPLINK_TO_AIRTEL_PRIMARY
 ip address 182.79.1.2 255.255.255.252
 ip nat outside
 ip virtual-reassembly in
!
interface GigabitEthernet0/3
 description UPLINK_TO_JIO_SECONDARY
 ip address 49.207.1.2 255.255.255.252
 ip nat outside
 ip virtual-reassembly in
!
no ip http server
no ip http secure-server
ip nat translation timeout 30
ip nat inside source route-map RM_NAT_AIRTEL interface
GigabitEthernet0/2 overload
ip nat inside source route-map RM_NAT_JIO interface
GigabitEthernet0/3 overload
!
ip route 0.0.0.0 0.0.0.0 182.79.1.1 track 1
ip route 0.0.0.0 0.0.0.0 49.207.1.1 10
ip route 8.8.4.4 255.255.255.255 182.79.1.1
ip route 10.10.10.0 255.255.255.0 172.16.16.2
ip route 10.10.10.0 255.255.255.0 172.16.17.2 250
ip route 10.10.11.0 255.255.255.0 172.16.16.2
ip route 10.10.11.0 255.255.255.0 172.16.17.2 250
ip route 10.10.12.0 255.255.255.0 172.16.16.2
ip route 10.10.12.0 255.255.255.0 172.16.17.2 250
ip route 10.10.13.0 255.255.255.0 172.16.16.2
ip route 10.10.13.0 255.255.255.0 172.16.17.2 250
ip route 10.10.14.0 255.255.255.0 172.16.16.2
ip route 10.10.14.0 255.255.255.0 172.16.17.2 250
ip route 10.10.15.0 255.255.255.0 172.16.16.2
ip route 10.10.15.0 255.255.255.0 172.16.17.2 250
! Under review, see Review notes item 1
ip route 0.0.0.0 0.0.0.0 8.8.8.8 track 1
ip route 0.0.0.0 0.0.0.0 4.4.4.4 10
ip route 1.1.1.1 255.255.255.255 8.8.8.8
!
ip access-list standard LAN_NETWORKS
 permit 10.10.0.0 0.0.255.255
!
ip sla 10
 icmp-echo 8.8.4.4 source-interface GigabitEthernet0/2
 frequency 5
ip sla schedule 10 life forever start-time now
!
route-map RM_NAT_JIO permit 10
 match ip address LAN_NETWORKS
 match interface GigabitEthernet0/3
!
route-map RM_NAT_AIRTEL permit 10
 match ip address LAN_NETWORKS
 match interface GigabitEthernet0/2
end
```

Core-01

Core switch, HSRP active, STP root

```
hostname Core-01
!
username anurag privilege 15 secret 5
$1$F3Rc$3hBm0G4nLTEGNvDfwhSye0
no aaa new-model
!
spanning-tree mode rapid-pvst
spanning-tree loopguard default
spanning-tree extend system-id
spanning-tree vlan 10-15 priority 4096
!
track 1 interface GigabitEthernet0/2 line-protocol
!
interface Port-channel1
 description Inter-Core_Trunk_Link
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface GigabitEthernet0/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-group 1 mode active
!
interface GigabitEthernet0/1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-group 1 mode active
!
interface GigabitEthernet0/2
 description Uplink to Router Gi0/0
 no switchport
 ip address 172.16.16.2 255.255.255.252
!
interface GigabitEthernet0/3
 description Downlink Trunk to DIST Gi0/0
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 999
 switchport mode trunk
 spanning-tree guard root
!
interface GigabitEthernet1/0
 description SPARE - not cabled
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 999
 switchport mode trunk
 spanning-tree guard root
!
interface Vlan10
 description Device Mgmt
 ip address 10.10.10.2 255.255.255.0
```

```
 standby 10 ip 10.10.10.1
 standby 10 priority 110
 standby 10 preempt
 standby 10 authentication md5 key-string SecGroup10
 standby 10 track 1 decrement 20
!
interface Vlan11
 description Data Vlan
 ip address 10.10.11.2 255.255.255.0
 standby 11 ip 10.10.11.1
 standby 11 priority 110
 standby 11 preempt
 standby 11 authentication md5 key-string SecGroup11
 standby 11 track 1 decrement 20
!
interface Vlan12
 description Voice
 ip address 10.10.12.2 255.255.255.0
 standby 12 ip 10.10.12.1
 standby 12 priority 110
 standby 12 preempt
 standby 12 authentication md5 key-string SecGroup12
 standby 12 track 1 decrement 20
!
interface Vlan13
 description GUEST
 ip address 10.10.13.2 255.255.255.0
 standby 13 ip 10.10.13.1
 standby 13 priority 110
 standby 13 preempt
 standby 13 authentication md5 key-string SecGroup13
 standby 13 track 1 decrement 20
!
interface Vlan14
 description BYOD
 ip address 10.10.14.2 255.255.255.0
 standby 14 ip 10.10.14.1
 standby 14 priority 110
 standby 14 preempt
 standby 14 authentication md5 key-string SecGroup14
 standby 14 track 1 decrement 20
!
interface Vlan15
 description Corentine Vlan
 ip address 10.10.15.2 255.255.255.0
 standby 15 ip 10.10.15.1
 standby 15 priority 110
 standby 15 preempt
 standby 15 authentication md5 key-string SecGroup15
 standby 15 track 1 decrement 20
!
ip http server
ip route 0.0.0.0 0.0.0.0 172.16.16.1
ip ssh time-out 60
ip ssh version 2
ip ssh server algorithm encryption aes128-ctr aes192-
ctr aes256-ctr
ip ssh client algorithm encryption aes128-ctr aes192-
ctr aes256-ctr
end
```

Core-02

Core switch, HSRP standby

```
hostname Core-02
!
username anurag privilege 15 secret 5
$1$k.Oj$.7Ve5STy3drgvYsg1waNp1
no aaa new-model
!
ip domain-name Hydent@1234
!
spanning-tree mode rapid-pvst
spanning-tree loopguard default
spanning-tree extend system-id
spanning-tree vlan 10-15 priority 8192
!
track 1 interface GigabitEthernet0/2 line-protocol
!
interface Port-channel1
 description Inter-Core_Trunk_Link
 switchport trunk encapsulation dot1q
 switchport mode trunk
!
interface GigabitEthernet0/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-group 1 mode active
!
interface GigabitEthernet0/1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-group 1 mode active
!
interface GigabitEthernet0/2
 description Uplink to Router Gi0/1
 no switchport
 ip address 172.16.17.2 255.255.255.252
!
interface GigabitEthernet0/3
 description Downlink Trunk to DIST Gi0/1
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 999
 switchport mode trunk
 spanning-tree guard root
!
interface GigabitEthernet1/0
 description SPARE - not cabled
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 999
 switchport mode trunk
 spanning-tree guard root
!
interface Vlan10
 description Device Mgmt
 ip address 10.10.10.3 255.255.255.0
 standby 10 ip 10.10.10.1
 standby 10 preempt
 standby 10 authentication md5 key-string SecGroup10
 standby 10 track 1 decrement 20
!
interface Vlan11
 description Data Vlan
 ip address 10.10.11.3 255.255.255.0
 standby 11 ip 10.10.11.1
 standby 11 preempt
 standby 11 authentication md5 key-string SecGroup11
 standby 11 track 1 decrement 20
!
interface Vlan12
 description Voice
 ip address 10.10.12.3 255.255.255.0
 standby 12 ip 10.10.12.1
 standby 12 preempt
 standby 12 authentication md5 key-string SecGroup12
 standby 12 track 1 decrement 20
!
interface Vlan13
 description GUEST
 ip address 10.10.13.3 255.255.255.0
 standby 13 ip 10.10.13.1
 standby 13 preempt
 standby 13 authentication md5 key-string SecGroup13
 standby 13 track 1 decrement 20
!
interface Vlan14
 description BYOD
 ip address 10.10.14.3 255.255.255.0
 standby 14 ip 10.10.14.1
 standby 14 preempt
 standby 14 authentication md5 key-string SecGroup14
 standby 14 track 1 decrement 20
!
interface Vlan15
```

```
 description Corentine Vlan
 ip address 10.10.15.3 255.255.255.0
 standby 15 ip 10.10.15.1
 standby 15 preempt
 standby 15 authentication md5 key-string SecGroup15
 standby 15 track 1 decrement 20
!
ip http server
ip route 0.0.0.0 0.0.0.0 172.16.17.1
ip ssh time-out 60
ip ssh version 2
ip ssh server algorithm encryption aes128-ctr aes192-
ctr aes256-ctr
ip ssh client algorithm encryption aes128-ctr aes192-
ctr aes256-ctr
end
```

DIST

Distribution switch, Layer 2

```
hostname DIST
!
spanning-tree mode rapid-pvst
spanning-tree portfast edge bpduguard default
spanning-tree extend system-id
!
interface GigabitEthernet0/0
 description Uplink Trunks to Cores
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 999
 switchport mode trunk
!
interface GigabitEthernet0/1
 description Uplink Trunks to Cores
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 999
 switchport mode trunk
!
interface GigabitEthernet0/2
 description Downlink Trunks to Access Layer
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 999
 switchport mode trunk
!
interface GigabitEthernet0/3
 description SPARE - not cabled
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 999
 switchport mode trunk
!
interface Vlan10
 description Management IP
 ip address 10.10.10.10 255.255.255.0
!
ip http server
ip route 0.0.0.0 0.0.0.0 10.10.10.1
ip ssh time-out 60
ip ssh version 2
ip ssh server algorithm encryption aes128-ctr aes192-
ctr aes256-ctr
ip ssh client algorithm encryption aes128-ctr aes192-
ctr aes256-ctr
end
```

Access_Finance

Access switch

```
hostname Access_Finance
!
username anurag privilege 15 secret 5
$1$kCJV$3ToVl73I/0ADo/w.M1oW6.
no aaa new-model
no ip domain-lookup
ip domain-name Hydent@1234
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
interface GigabitEthernet0/0
 description Uplink Trunks to Distribution
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 999
 switchport mode trunk
!
interface GigabitEthernet0/1
 description Host Endpoint Connection
 switchport access vlan 11
 switchport mode access
 switchport port-security mac-address sticky
 switchport port-security mac-address sticky
0050.7966.6808
 switchport port-security
 spanning-tree portfast edge
 spanning-tree bpduguard enable
!
interface GigabitEthernet0/2
 description SPARE - not cabled
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 999
 switchport mode trunk
!
interface Vlan10
 description Management IP
 ip address 10.10.10.15 255.255.255.0
!
ip http server
ip route 0.0.0.0 0.0.0.0 10.10.10.1
ip ssh time-out 60
ip ssh version 2
ip ssh server algorithm encryption aes128-ctr aes192-
ctr aes256-ctr
ip ssh client algorithm encryption aes128-ctr aes192-
ctr aes256-ctr
end
```

Airtel

Primary ISP (simulated)

```
hostname Airtel
!
interface GigabitEthernet0/0
 description FACING_EDGE_ROUTER
 ip address 182.79.1.1 255.255.255.252
!
interface GigabitEthernet0/1
 description FACING_GOOGLE_SERVER
 ip address 182.79.2.1 255.255.255.252
!
no ip http server
no ip http secure-server
ip route 8.8.4.4 255.255.255.255 182.79.2.2
ip route 8.8.8.8 255.255.255.255 182.79.2.2
end
```

JIO

Secondary ISP (simulated)

```
hostname JIO
!
interface GigabitEthernet0/0
 description FACING_EDGE_ROUTER
 ip address 49.207.1.1 255.255.255.252
!
interface GigabitEthernet0/1
 description FACING_GOOGLE_SERVER
 ip address 49.207.2.1 255.255.255.252
!
no ip http server
no ip http secure-server
ip route 8.8.8.8 255.255.255.255 49.207.2.2
end
```

Google_server

Internet host (simulated)

```
hostname Google_server
!
interface Loopback0
 ip address 8.8.8.8 255.255.255.255
!
interface Loopback1
 ip address 8.8.4.4 255.255.255.255
!
interface GigabitEthernet0/0
 description LINK_TO_AIRTEL
 ip address 182.79.2.2 255.255.255.252
!
interface GigabitEthernet0/1
 description LINK_TO_JIO
 ip address 49.207.2.2 255.255.255.252
!
no ip http server
no ip http secure-server
ip route 49.207.1.0 255.255.255.252 49.207.2.1
ip route 182.79.1.0 255.255.255.252 182.79.2.1
end
```